

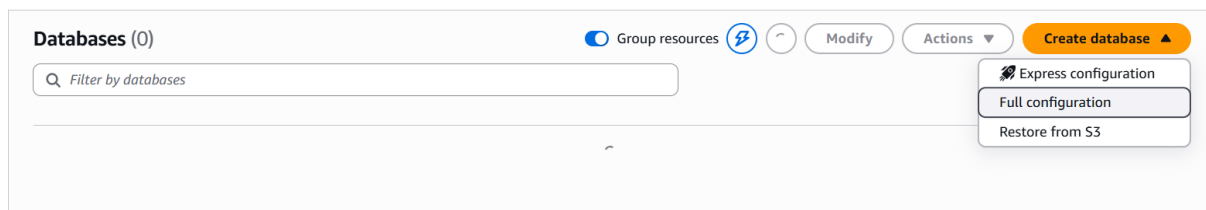
Amazon RDS for MySQL Zero-ETL Integration with Amazon Redshift

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to configure an **Amazon RDS for MySQL Zero-ETL integration with Amazon Redshift Serverless**. First, an RDS MySQL instance is created, connected using MySQL Workbench, and populated with a sample database containing employee and department tables. Next, an Amazon Redshift Serverless workgroup and namespace are created. A Zero-ETL integration is then configured between the RDS instance and the Redshift namespace, with AWS automatically applying the required settings. Although the integration becomes active, the destination database is not created automatically, preventing tables from appearing in Redshift. To resolve this, a destination database is manually created from the integration page in Redshift. After the database is provisioned and activated, the synchronized MySQL database and tables become available in the Redshift Query Editor, enabling near real-time analytics without building ETL pipelines.

- Open Amazon RDS
- Login to AWS Console
- Search for RDS
- Open Amazon RDS
- Click Create Database
- Choose Full Configuration



- Choose Database Creation Method
- Select:
- Standard Create
- Select Engine
- Choose:
- Engine Type: MySQL
- Version:
- MySQL 8.0.x
- (Use the latest supported version)

Create database [Info](#)

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle

ORACLE

☐ Microsoft SQL Server



☐ IBM Db2

IBM Db2

Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

- Select Template
- Choose:
- Free Tier
- (or Dev/Test if Free Tier is unavailable)

Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

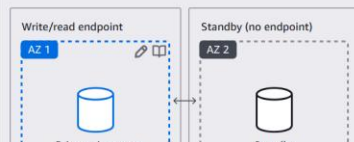
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



☐ Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

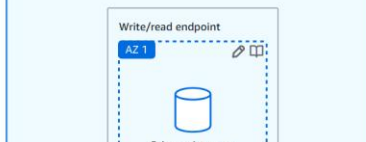
- 99.95% uptime
- Redundancy across Availability Zones



☒ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



- Configure DB Instance
- DB Instance Identifier
 - mysql-zero-etl
- Master Username
 - Admin
- Choose Self-Managed
- Master Password
- Confirm Password

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

mysql-zero-etl-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

- Instance Configuration
- Choose Burstable classes
- Choose:
 - db.t3.micro
- (or any supported instance class)

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☒ **Burstable classes (includes t classes)**

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

- Storage Configuration
- Storage Type
 - General Purpose SSD (gp3)
- Allocated Storage
 - 20 GB

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

20 GiB

Allocated storage value must be 20 GiB to 6,144 GiB

► Additional storage configuration

- Connectivity
- VPC
- Select:
- Default VPC

- or your custom VPC.

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ **Dual-stack mode**
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-062d767592c9ad3e0) [Info](#)
2 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

☐ After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-062d767592c9ad3e0
2 Subnets, 2 Availability Zones

Public access [Info](#)

- Public Access
- For MySQL Workbench connection:
- Yes
- VPC Security Group
- Create new:
 - mysql-zero-etl-sg
- or use existing security group.
- Database Port
 - 3306

Public access [Info](#)

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups
Choose one or more options

mysql-zero-etl-sg [X](#)

Availability Zone [Info](#)
No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

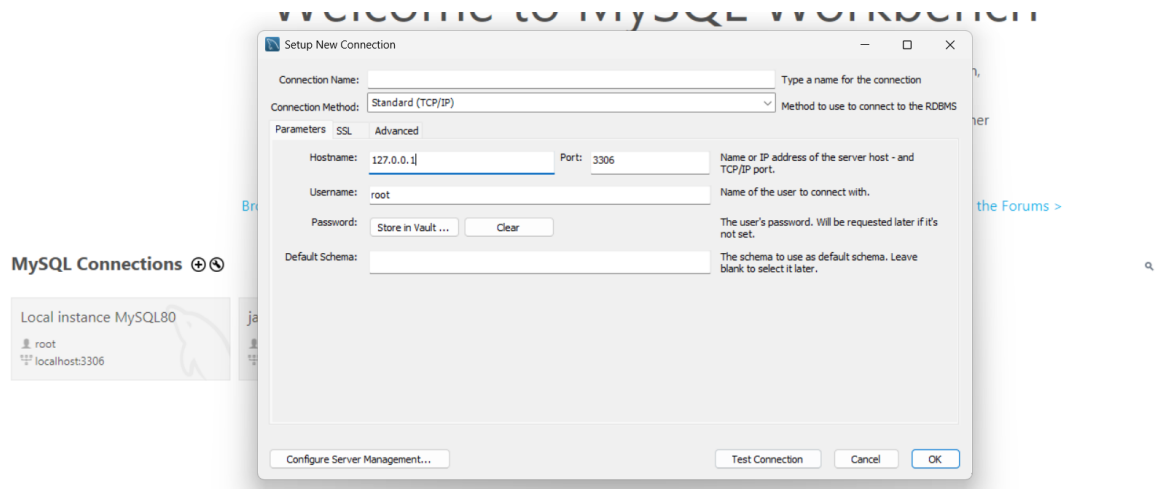
☐ **Create an RDS Proxy** [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

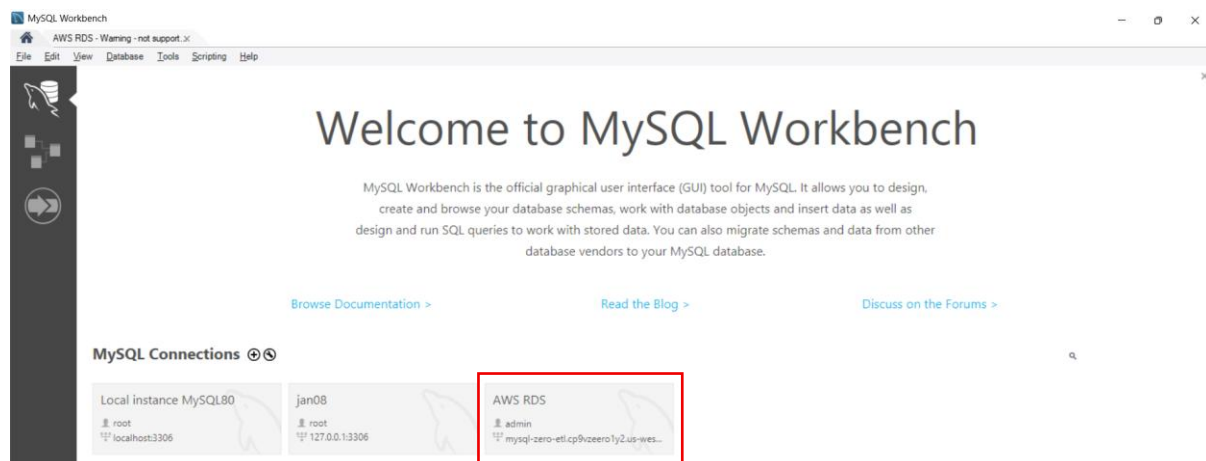
rds-ca-rsa2048-g1 (default)

- Click:
 - Create Database
- Wait for:
- Status = Available
- Connect Using MySQL Workbench
- Use:
 - Hostname = RDS Endpoint

- Port = 3306
- Username = admin
- Password = Your Password



- Test Connection → OK



- Create Sample Database
- Run:

```
CREATE DATABASE companydb;
USE companydb;
```

- Step 4: Create Tables
- Employees Table

```
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100),
```

```
department VARCHAR(50),  
salary DECIMAL(10,2)  
);
```

- Departments Table

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(100)  
);
```

- Step 5: Insert Sample Data
- Employees

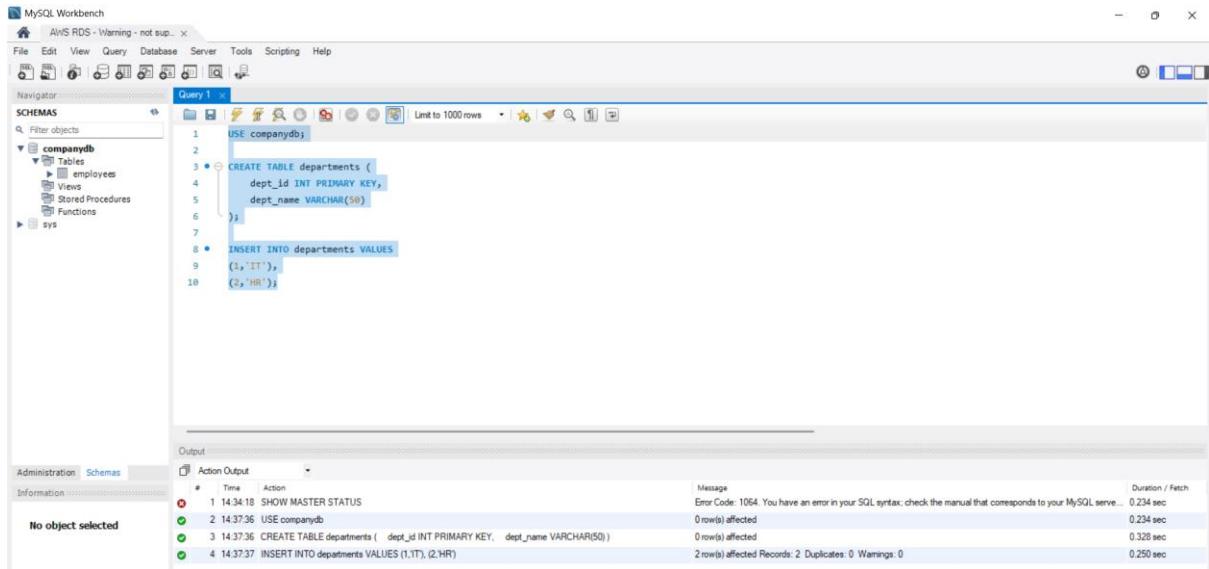
```
INSERT INTO employees  
VALUES  
(101,'Aman','IT',50000),  
(102,'Rahul','HR',45000),  
(103,'Priya','Finance',60000);
```

- Departments

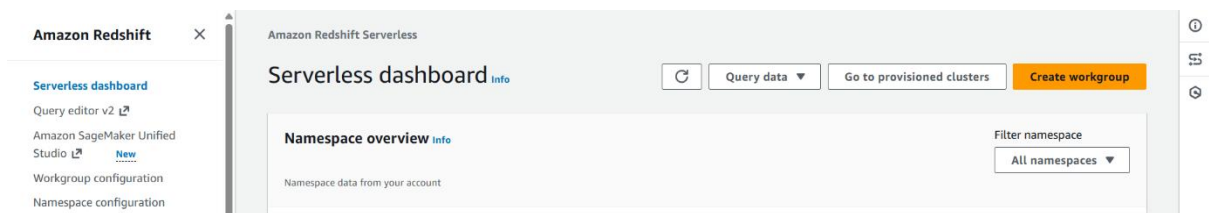
```
INSERT INTO departments  
VALUES  
(1,'IT'),  
(2,'HR'),  
(3,'Finance');
```

- Verify:

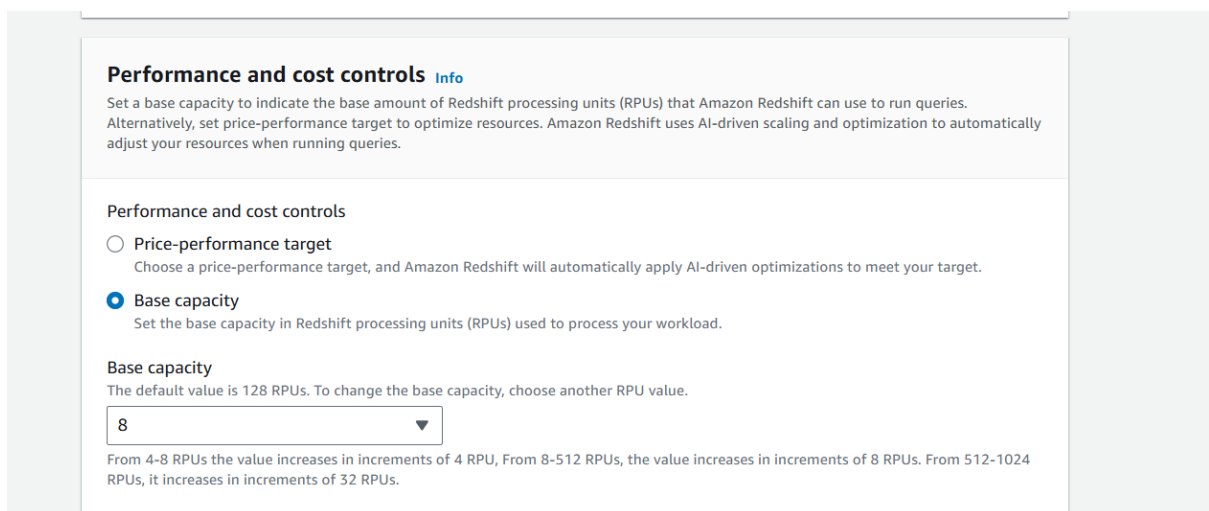
```
SELECT * FROM employees;  
SELECT * FROM departments;
```



- Create Redshift Serverless
- Open:
- AWS Console → Amazon Redshift → Serverless → Create Workgroup



- Configure Workgroup
- Workgroup Name
 - zero-etl-workgroup
- Base Capacity
 - Selected:
 - 8 RPUs



- Configure Namespace

- Namespace Name
 - zero-etl-namespace

Step 2
Choose namespace

Step 3
Review and create

Namespace
Namespace is a collection of database objects and users. Data properties include database name and password, permissions, and encryption and security.

☒ Create a new namespace
☐ Add to an existing namespace

Namespace
This is a unique name that defines the namespace.

The name must be from 3-64 characters. Valid characters are a-z (lowercase only), 0-9 (numbers), and - (hyphen).

- Admin Username
 - admin
- Admin Password
 - <Your Password>
- Database Name
 - dev
- Create.
- Wait until:
- Status = Available

Admin user credentials
IAM credentials provided as your default admin user credentials. To add a new admin username and password, customize admin user credentials.

☒ Customize admin user credentials
To use the default IAM credentials, clear this option.

Admin user name
The administrator's user name for the first database.

The name must be 1-128 alphanumeric characters, and it can't be a reserved word.

Admin password
Select an option to manage your admin password.

☒ Manage admin credentials in AWS Secrets Manager [Info](#)
 AWS manages a KMS key that encrypts your data.
 Your data is encrypted by default with a key that AWS owns and manages for you. To choose a different key, customize your encryption settings.

☐ Customize encryption settings (advanced)

☐ Generate a password
 Amazon Redshift generates an admin password.

☐ Manually add the admin password
 Manually enter the admin password.

- Create Zero-ETL Integration
- Open:
- mysql-zero-etl → Zero-ETL Integrations → Create Integration

Monitoring | Logs & events | Configuration | **Zero-ETL integrations** | Maintenance & backups | Data migrations | Tags

Zero-ETL integrations (0) [Info](#)

View the integrations in your account.

Integration name ▲	Status ▼	Creation date ▼	Source ARN ▼	Destination AWS account ▼	Destination type ▼	Destination
No integrations						

- Integration Name
 - mysql-redshift-zero-etl

Step 1: Getting started (selected)

Step 2: Select source

Step 3: Select target

Step 4 - optional: Add tags and encryption

Step 5: Review and create

Getting started

Integration identifier
Enter a name for your integration. The name must be unique across all integrations owned by your AWS account in the current AWS Region.

Integration identifier:

1-63 alphanumeric characters or hyphens. The first character must be a letter.

Integration description - optional
Enter a description for your integration.

[Cancel](#) [Next](#)

- Source
 - mysql-zero-etl
- Check the Fix it for me
- Click Reboot and continue

Source reboot required

In order to create and apply the required parameters for a zero-ETL integration, a reboot is required. This will happen immediately after choosing **Reboot and continue**. To schedule the reboot during a maintenance window, update the parameter settings through the normal process in the RDS console.

RDS will create a new parameter group based on the current parameter group that's associated with the database, then make the following changes:

Group name	Parameter group family
zero-etl-params-1781940665641	mysql8.4

New parameters set	
binlog_format = ROW	binlog_row_image = full

[Cancel](#) [Reboot and continue](#)

- Click Next

Aurora and RDS > Zero-ETL integrations > Create zero-ETL integration

Step 1: Getting started
Step 2: **Select source**
Step 3: Select target
Step 4 - optional: Add tags and encryption
Step 5: Review and create

Select source Info

Select a source database to replicate data into the target resource. Specific parameter values must be set in the source DB parameter group in order to create an integration. RDS can configure them for you during setup, or you can configure them manually in the target service.

Source

Source database
The source database where the data is replicated from. Only databases running the supported versions are available.

mysql-zero-etl 🔄 Browse RDS databases

⚠️ Fix parameter values
Certain parameter settings are required to successfully create an integration. You can have RDS fix the parameter values and reboot the database for you, or you can manually update them in [parameter group settings](#).

☐ Fix it for me (requires reboot)
RDS will create a new parameter group based on the parameter group that's currently associated with the chosen database, then it will apply the correct parameter values.

Data filtering options - optional Info
Include or exclude any existing and future database table that matches your entered list of filter expressions. All tables are included by default.

☐ Customize data filtering options

Cancel Previous Next

- Target
- Amazon Redshift Data warehouse

Aurora and RDS > Zero-ETL integrations > Create zero-ETL integration

✔ Successfully rebooted mysql-zero-etl database and applied parameter settings. View details ✕

Step 1: Getting started
Step 2: Select source
Step 3: **Select target**
Step 4 - optional: Add tags and encryption
Step 5: Review and create

Select target Info

Select the target resources type. Before you create an integration, you must specify authorized principals and integration source on the target data warehouse, and enable case sensitivity. RDS can complete these steps for you during setup, or you can configure them manually in the target service.

Target Info

AWS account
Choose if the target resource is in the current AWS account or a different account.

☒ Use the current account
☐ Specify a different account

Target resource type
Choose the target resource type.

☒ Amazon Redshift data warehouse
☐ AWS Glue Catalog

Amazon Redshift data warehouse
Only encrypted data warehouses are available. To create a new Redshift data warehouse or encrypt an existing one go to [Redshift console](#).

Browse Redshift data warehouses and select a target 🔄 Browse Redshift data warehouses

Cancel Previous Next

- Select:
- Namespace:
 - zero-etl-namespace

Choose a Redshift data warehouse ✕

Data warehouses (2) 🔄

Choose the Amazon Redshift data warehouse that will be the target for replicated data. Only encrypted data warehouses are available.

Target data warehouse	Target type	Status
☐ default-namespace	Serverless	Available
☒ zero-etl-namespace	Serverless	Available

Cancel Choose

Browse Redshift data warehouses and select a target 🔄 Browse Redshift data warehouses

- Check Fix it for me

[Aurora and RDS](#) > [Zero-ETL integrations](#) > Create zero-ETL integration

Getting started
Step 2
Select source
Step 3
Select target
Step 4 - optional
Add tags and encryption
Step 5
Review and create

Select target Info

Select the target resources type. Before you create an integration, you must specify authorized principals and integration source on the target data warehouse, and enable case sensitivity. RDS can complete these steps for you during setup, or you can configure them manually in the target service.

Target Info

AWS account
Choose if the target resource is in the current AWS account or a different account.

☒ Use the current account
☐ Specify a different account

Target resource type
Choose the target resource type.

☒ Amazon Redshift data warehouse
☐ AWS Glue Catalog

Amazon Redshift data warehouse
Only encrypted data warehouses are available. To create a new Redshift data warehouse or encrypt an existing one go to [Redshift console](#).

zero-etl-namespaces [Browse Redshift data warehouses](#)

☒ **Fix resource policy and case sensitivity parameter**
The selected target does not have the correct resource policy and case sensitivity parameter to support a zero-ETL integration. You can have RDS fix this for you, or you can manually update them in the Redshift console. [Learn more](#)

☐ **Fix it for me**
RDS will fix this by adding the current account and the selected source database in the resource policy and enabling case sensitivity.

- Click Continue

Review changes ✕

The modified parameters and resource policy are displayed here. These changes will be applied immediately after you choose **Continue**.

Resource policy
RDS will update the target data warehouse resource policy to have the following authorized principal and integration source.

Authorized principal 463646775279	Authorized integration source arn:aws:rds:us-west-2:463646775279:db:mysql-zero-etl
---	--

Parameters
RDS will enable case sensitivity for the target data warehouse.

Parameter name	Value
enable_case_sensitive_identifier	true

[Cancel](#) [Continue](#)

- Create Integration.

[Aurora and RDS](#) > [Zero-ETL integrations](#) > Create zero-ETL integration

Resource policy

Authorized principal
463646775279

Authorized integration source
arn:aws:rds:us-west-2:463646775279:db:mysql-zero-etl

Parameters

Parameter name	Value
enable_case_sensitive_identifier	true

Step 4: Add tags and encryption [Edit](#)

Tags

No tags have been added.

Encryption

Encryption settings AWS default key	AWS KMS key AWS owned key
--	------------------------------

- Wait For Integration
- Initially:
 - Status = Creating
- Wait until:
 - Status = Active

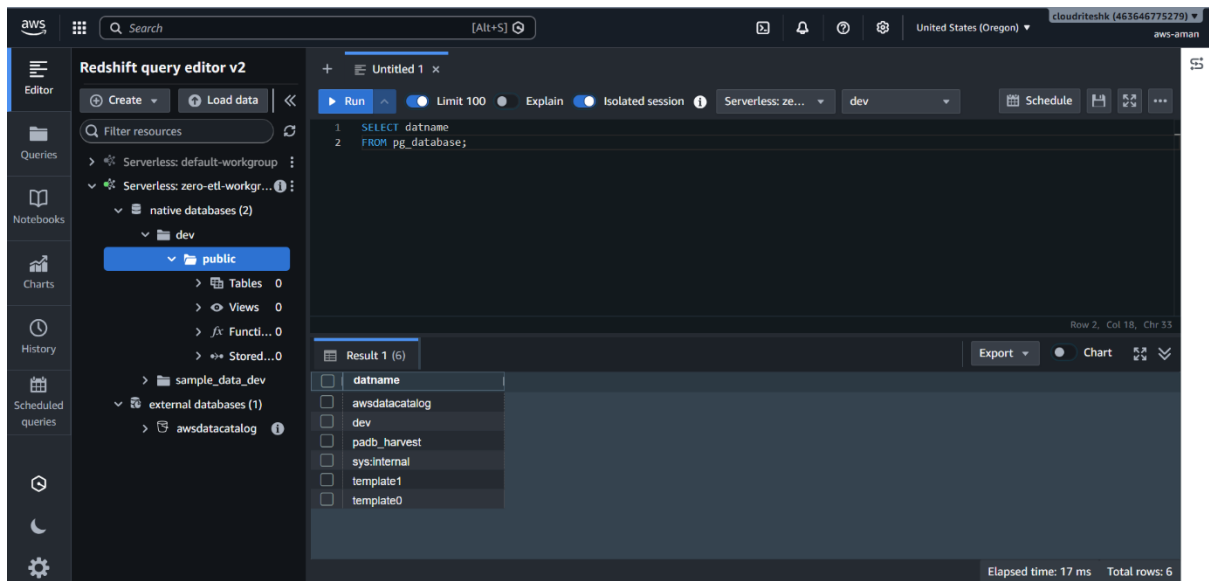
Zero-ETL integrations (1) [Info](#)
[Refresh](#)
[Modify](#)
[Delete](#)
[Create zero-ETL integration](#)

View the integrations in your account.

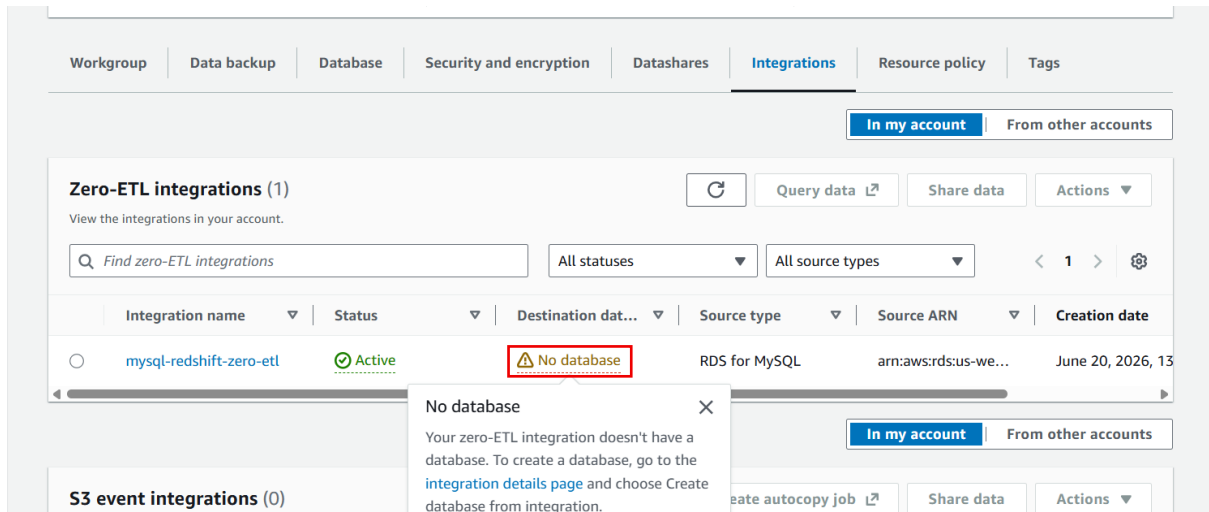
< 1 >

	Integration name	Status	Creation date	Source ARN	Dest
☐	mysql-redshift-zero-etl	Active	Jun 20, 2026, 13:09:07 (UTC+05:30)	arn:aws:rds:us-west-2:463646775279:db:mysql-zero-etl	4636

- First Problem We Faced
- Integration became:
 - Active
- But in Redshift:
 - No tables
 - No schemas
 - No database



- Root Cause
- Integration was active but destination database was not created.
- Integration page showed:
- Destination database:
 - No database
- and
- Database creation required



Introducing history mode for zero-ETL integrations
History mode keeps track in the integration target which records in the integration source were deleted or changed. [Learn more](#)

Amazon Redshift > Zero-ETL integrations > 5c577dab-4e6a-4ee1-90e5-129d5fb654dc

mysql-redshift-zero-etl Info

[Manage history mode](#) [Query data](#) [Share data](#)

Database creation required
To activate your integrations and access the data, you need to create a database for each integration. To create a database for the integration, choose Create database from integration.

Create database from integration

Zero-ETL integration details

General settings	Source	Target
Integration name mysql-redshift-zero-etl	Source type RDS for MySQL	Target type Redshift Serverless
Integration ID 5c577dab-4e6a-4ee1-90e5-129d5fb654dc	Source ARN arn:aws:rds:us-west-2:463646775279:db:mysql-zero-etl	Target ARN arn:aws:redshift-serverless:us-west-2:463646775279:namespace/e9ac0001-aa1e-479e-9518-e708ff199f78
Integration ARN arn:aws:rds:us-west-2:463646775279:integration:5c577dab-4e6a-4ee1-90e5-129d5fb654dc		Data warehouse name arn:aws:redshift-serverless:us-west-2:463646775279:namespace/e9ac0001-aa1e-479e-9518-e708ff199f78

- To fix this, Database creation required
- Create Database from Integration
- Open:
- Redshift → Zero-ETL Integrations → mysql-redshift-zero-etl
- Click:
- Create database from integration

Introducing history mode for zero-ETL integrations
History mode keeps track in the integration target which records in the integration source were deleted or changed. [Learn more](#)

Amazon Redshift > Zero-ETL integrations > 5c577dab-4e6a-4ee1-90e5-129d5fb654dc

mysql-redshift-zero-etl Info

[Manage history mode](#) [Query data](#) [Share data](#)

Database creation required
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Create database from integration

Zero-ETL integration details

General settings	Source	Target
Integration name mysql-redshift-zero-etl	Source type RDS for MySQL	Target type Redshift Serverless
Integration ID 5c577dab-4e6a-4ee1-90e5-129d5fb654dc	Source ARN arn:aws:rds:us-west-2:463646775279:db:mysql-zero-etl	Target ARN arn:aws:redshift-serverless:us-west-2:463646775279:namespace/e9ac0001-aa1e-479e-9518-e708ff199f78
Integration ARN arn:aws:rds:us-west-2:463646775279:integration:5c577dab-4e6a-4ee1-90e5-129d5fb654dc		Data warehouse name arn:aws:redshift-serverless:us-west-2:463646775279:namespace/e9ac0001-aa1e-479e-9518-e708ff199f78

- Enter:
- Destination Database Name:
 - **mysql_zero_etl**
- Click:

- Create Database

Create database from integration

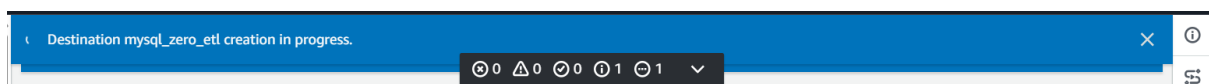
To start querying data in the integration, create a database from the integration.

Integration ID
5c577dab-4e6a-4ee1-90e5-129d5fb654dc

Data warehouse name
zero-etl-namespace

Destination database name
Enter a new name for your database.

- Started creating, it will take 10 to 15 min.



Once it is done, Destination database will be activated.

Integration name	Status	Destination database	Source type	Source ARN	Creation date
mysql-redshift-zero-etl	Active	Active	RDS for MySQL	arn:aws:rds:us-west-2:4...	June 20, 2026, 13:09 (UTC+05)

- Now go back to query editor, you can see the database there with the tables.

Editor

Queries

Notebooks

Charts

History

Scheduled queries

Redshift query editor v2

CreateLoad data

Filter resources

Serverless: default-workgroup

Serverless: zero-etl-workgroup

native databases (3)

dev

mysql_zero_etl

companydb

Tables2

Views0

Functions0

Stored procedures0

public

sample_data_dev

external databases (1)

Untitled 1

RunLimit 100ExplainIsolated sessionServerless: ze...mysql_zero_etl

Schedule

Filter...

dev

mysql_zero_etl

awsdatacatalog

1SELECT table_schema,
2table_name
3FROM information_schema.tables
4ORDER BY table_schema, table_name;

Row 4, Col 35, Chr 107

Result 1 (100)

ExportChart

table_schema	table_name
companydb	departments
companydb	employees
information_schema	applicable_roles
information_schema	check_constraints
information_schema	column_domain_usage
information_schema	column_privileges
information_schema	column_udt_usage
information_schema	columns
information_schema	constraint_column_usage

Query ID: 613329Elapsed time: 2046 msTotal rows: 100